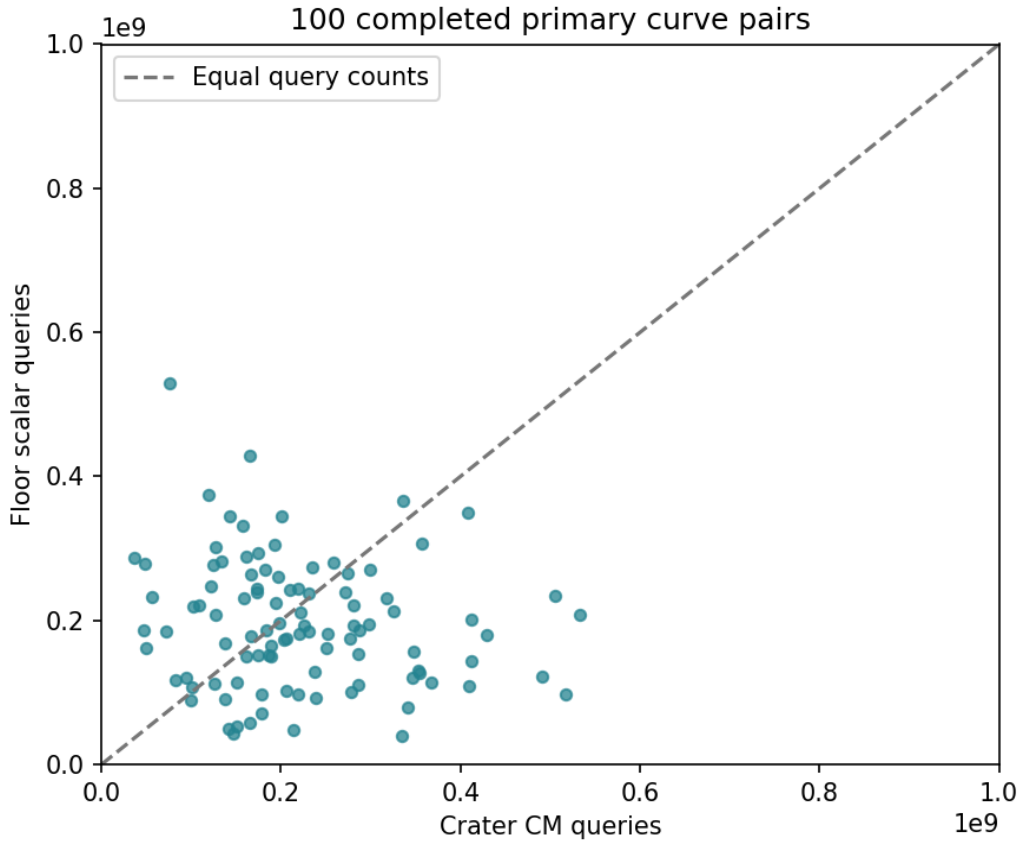
Conductor-gap research: completed native campaigns

15 October 2026 · Exact recoveries, structural certificates and remaining work

No systematic crater/floor mean timing difference is resolved in the 100-pair D = −19 population. A cheap CM endomorphism is concrete, while a generic recurrence gives the floor most of the same orbit-construction benefit. Intrinsic ECDLP hardness remains unestablished.

Completed experiment	Verified attempts / result
100-pair primary campaign	500 / 500: 300 folded IC and 200 same-point rho
Replication in 30 sessions	265 / 265: 155 IC and 110 rho
Same-crater map/scalar comparisons	145 exact agreements: 100 primary plus 45 replication
Independent replay	473 IC base/relation/matrix replays; 797 PARI target recoveries
Orbit construction	44,000 trials, including A/A, recurrence, pinned repeats and fresh-process batches

Primary floor/crater native-backend CPU: 0.980, paired 95% interval [0.866, 1.111]. Query ratio: 0.878 [0.764, 1.009]. Ten thousand whole-gap bootstrap resamples; all 100 endpoint pairs included.

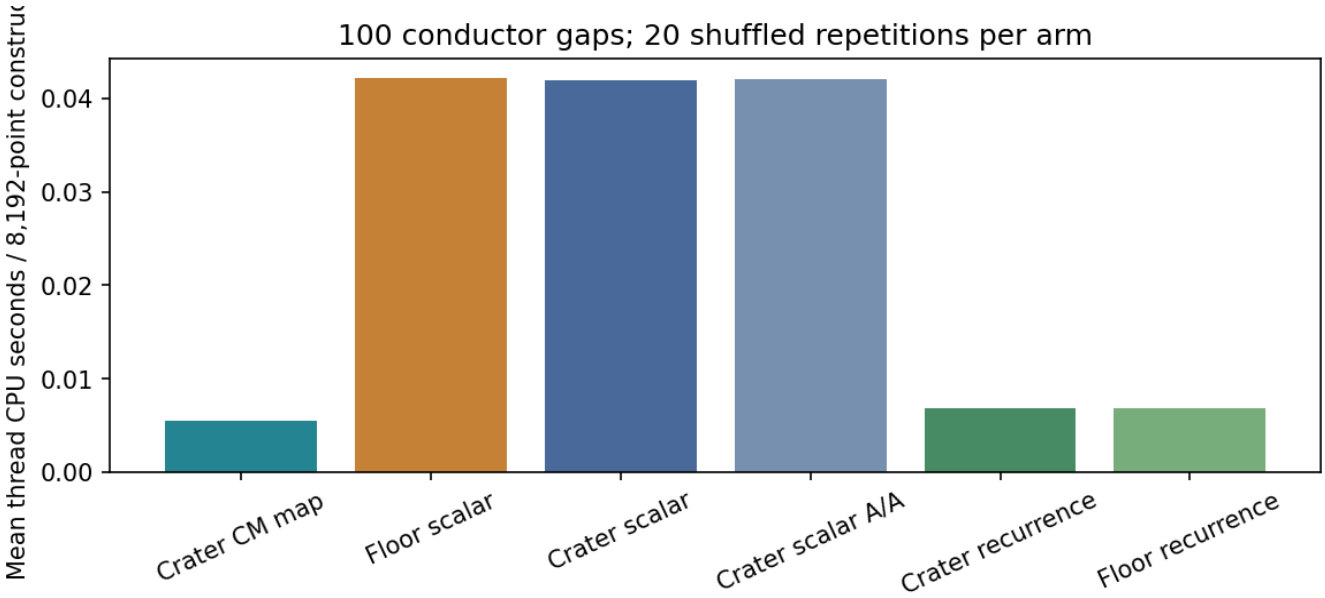


CPU observations are L0 diagnostics; endpoint strata ran mostly in separate queue waves. Bootstrap intervals do not remove host/wave confounding. Native-backend CPU includes orbit/table setup and native recovery. The measured process also includes parameter derivation and independent replay. Native prices remain incomplete; no complete-price IC/rho ratio or S=0 result is admitted.

1. The structural effect and its generic control

The tested pair family has $p = 5\ell^2 - \ell + 1$, $t = \ell - 2$, prime $N = 5\ell^2 - 2\ell + 4$, $D_K = -19$ and endpoint conductors 1 and ℓ. The 100 distinct prime gaps span 33,555,419–35,282,963; completed ECDLP comparisons use 53-bit fields.

On the crater a rational degree-5 CM map acts as [λ], where $\lambda^2 - \lambda + 5 = 0$ modulo N. The floor order has no degree-5 endomorphism. Nevertheless $P[k+2] = P[k+1] - [5]P[k]$, after one initial [λ] multiplication per seed, generates the same linked orbit with generic arithmetic on either endpoint.



CPU 0 pinned; local monitor uncontended. Physical-host/frequency isolation unverified.

Pinned means per 8,192-point construction: crater CM 5.464 ms; crater recurrence 6.859 ms; floor recurrence 6.857 ms; direct floor scalar multiplication 42.141 ms. The local monitor reported no contention: 0.34 other CPU seconds during 300.43 wall seconds. Physical-host and frequency isolation are unverified.

Twenty fresh-process batches add 12,000 construction records. Each batch contains all 600 model/arm jobs and uses precomputed CM parameters; it is not a separate process per arm. These batches ran at L0. First observed passes and later shuffled passes are recorded separately. Cold/warm cohorts are not a causal timing comparison.

Four public seed logs and 2,048 levels yield 8,192 factor-base points. Signed pair sums produce equations in four seed logs and the target. Scalar-proportional column parts are rejected; the matrix may pin the target before determining every seed log. Independent replay checks the base, each relation, rank and $[d]G = Q$.

For 136 cases, independent PARI point-order and Weil-pairing witnesses certify crater-twist invariant factors ℓ and $q\ell$, versus a cyclic floor twist of order $q\ell^2$ ($q = 5$ or 11). This gives derived rational degree-ℓ kernel-line counts $\ell+1$ versus one. These counts are not a complete isogeny enumeration. Both original ECDLP endpoints retain the same prime order N.

2. Replication and the current curve

Cohort / gap clusters	Query ratio, floor/crater	Native CPU ratio, floor/crater
fresh / 10	0.988 [0.845, 1.149]	1.059 [0.912, 1.221]
isomorphic / 10	1.397 [0.882, 2.056]	1.465 [0.974, 2.049]
d43 / 10	1.375 [0.998, 2.011]	1.339 [1.006, 1.887]
pilot-five-rounds / 1	0.977 (descriptive)	0.951 (descriptive)

Intervals use 10,000 whole-gap cluster resamples, retaining all targets/rounds within each sampled gap. The five-round pilot has only one gap and therefore no curve-population interval. Twenty additional floor vertices completed another 40 IC/rho attempts.

The D = −43 native CPU interval narrowly excludes one in this ten-pair exploratory subset; the excess CPU accompanies extra queries. Intervals are unadjusted for multiple comparisons, and timings are L0. This is a replication lead, not a general conductor-hardness conclusion. The coordinate control transports G and Q but reruns prefix seed selection, so it also changes geometric seed points.

The completed 200-target rho panel has a posthoc floor/crater full-process CPU ratio 1.174 [0.99897, 1.38318], with mean CPU 14.335s / 16.834s. Its charged group-work lower-bound ratio is 1.160 [0.986, 1.367]. Both intervals include one. Canonicalisation remains unpriced; endpoint waves confound CPU. These within-rho diagnostics are not an admitted IC/rho speed comparison.

Pair 46 parameter	Exact value
p	5947698030651343
N = r; cofactor	5947697996161643; 1
t; prime gap ℓ	34489701; 34489703
Crater $y^2 = x^3 + ax + b$	a = 5757116383569871; b = 3560173328417442
Floor $y^2 = x^3 + ax + b$	a = 2087370704375809; b = 1402887937802154

The generic-recurrence pilot recovered both pair-46 targets and reproduced the direct-scalar factor-base hashes and query counts exactly. Across all completed sessions, 797 target recoveries passed independent PARI replay, including prime/Hasse witnesses and rejected scalar mutations. All 473 IC transcripts passed native public-base, relation and rank replay, including corrupted-sign rejection.

Six initial small-control preflight failures exposed loss of public fixtures in measured-child normalization. CurveSpec::for_child fixes that transport. Failed receipts remain preserved. The full Rust library run passed 4,116 tests (117 ignored); current fixture and CM-map checks and pinned Rust 1.98 strict Clippy also pass. Conditional Lean recurrence/orbit/norm lemmas are axiom-free; CM classification and point-map bridges are not claimed as formalized.

3. Larger gaps, scope and next steps

The fast codomain constructor matched all 100 direct references; the batched variant also passed forced small-batch comparisons. Independent PARI checks certify 15 new floor models through 91 bits and 20 extra floor vertices. The largest independently certified gap in this snapshot is 17,592,186,219,863 over p = 154742507997167055469247398 3.

The 6 GiB upper-limit attempts hit wall or allocation caps. The 11 GiB retry reached 5,242,880 of 5,590,170 giant points before its 3,600-second cap. The 14 GiB precomputed-tree attempt was OOM-killed after 939.36 seconds when concurrent test compilation grew; that scheduling failure is preserved. A new optimized bounded-batch variant passes 4,096 coefficient checks and both 100-reference campaigns. Its guarded retry is running with an 11 GiB cap and 5,400 seconds, after local compilers exited. A floor counts only after independent certification; point-map and dual-map certification remain separate.

The older near-limit candidate, ℓ = 129149576694539 and p = 792281625117984032632351147 31, is separate. Its nonzero kernel point needs extension degree 2,226,716,839,561 (twist point degree 4,453,433,679,122). A dense 96-bit basis would need about 26.7 TB per field element. This is a prerequisite barrier for the current rational-point method; other kernel representations are not excluded. Its floor is unconstructed.

The earlier depth-20 degree-2 volcano remains a separate completed export: 21 levels, 1,048,576 explicit chosen-trace models and all intermediate strata. It is a smooth conductor ratio traversed by small steps. The large-prime benchmark samples endpoints and 20 additional floors, not every vertex of every prime-degree volcano. Earlier prefix-IC censored results are preserved and are not converted into solve times.

• Repeat D = −43 on more fresh targets and gaps, with the generic degree-11 recurrence control; preserve failures and use curve-level clustering.

• Transport the four geometric seed points in coordinate controls, as well as G and Q, to isolate arithmetic from prefix-seed selection.

• Price all native components before a complete IC/rho work claim. Retain cold setup and all five exclusive online phases, including independent replay; earn isolation for primary wall-time comparisons.

• Complete and independently certify the expanded 96-bit construction. The original non-rational-kernel candidate needs a compatible kernel representation.

• A prime-field Semaev/F4/SAT complexity claim requires a separately implemented and independently checked backend. The completed folded MITM experiment makes no such claim.

Sources: REPORT.md, native source, frozen input/binary manifests, results/paired-primary.json, results/replication-metrics.json, results/isolated-summary.json, replay and PARI certificates, and per-session reports. The original six-page summary and evidence ledger remain in inputs/historical-evidence-summary.zip. Rebuild: python native/render_report.py (rendering only). PR: github.com/aburan28/crypto/pull/1888.